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COVID-19 disrupted patterns of cause-specific mortality in Switzerland

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Background

- The COVID-19 pandemic reshaped mortality through:
 - Direct deaths from SARS-CoV-2,
 - Associated deaths from other causes (e.g., cardiovascular diseases) triggered by infection,
 - Mortality displacement across time and causes,
 - Indirect effects of control measures and behavioural changes (e.g., fewer respiratory infections, healthcare disruptions).
- These disruptions occur simultaneously across multiple causes, limiting the interpretability of all-cause and single-cause excess mortality.
- A multivariate approach is necessary to capture cross-cause dependencies and facilitate interpretation of these disruptions.

Objectives

- Implement a Bayesian multivariate model to estimate cause-specific excess mortality over time in Switzerland.
- Quantify dependencies between mortality causes before and during COVID-19.
- Identify the mechanisms driving mortality disruptions during the pandemic.

Data

- Cause-specific deaths:** $y_{irs}(t)$
 - aggregated by cause i , sex s , NUTS-2 region r , year and week t ; by age group (0–17, 18–39, 40–64, 65–79, 80+); in 2011–2021,
 - include immediate, underlying, and contributing causes (ICD-10),
 - grouped into 9 categories: cardiovascular (CVD), mental/neurological, respiratory, infectious diseases, cancer, external causes, suicide, COVID-19 and others.
- Population sizes:** N_{rst} by age, sex, region; weekly estimates via linear interpolation.

Model

- Expected mortality:** For each age group, we fit the number of deaths $y_{irs}(t)$ due to underlying cause i at year and week t , in region r and in sex s between 2011 and 2019:

$$y_{irs}(t) \sim \text{Poisson}(\mu_{irs}(t))$$

with $\log \mu_{irs}(t) = \alpha_i + \beta_{ir} + \gamma_{is} + f_i(t) + g_i(t) + \varepsilon_i(t) + \log N_{rst}$ and where

- $\alpha_i, \beta_{ir}, \gamma_{is}$: baseline mortality, effects of region and sex,
- $f_i(t), g_i(t)$: long-term and seasonal trends (Gaussian processes),
- $\varepsilon_i(t)$: multivariate random effect, $\varepsilon(t) \sim \mathcal{N}(0, \Sigma)$.

- Excess mortality:**

$$E_{irs}(t) := y_{irs}(t) - \tilde{y}_{irs}(t).$$

where $\tilde{y}_{irs}(t)$ is the expected number of deaths (i.e., posterior predictive distribution after removing $\varepsilon_i(t)$).

Temporal trends and cross-cause correlations

- The model captures seasonal and long-term trends of cause-specific mortality in 2011–2019 (see Fig 1 for 80+).
- Expected mortality peaks between January and February for every cause.
- Shift in peak timing during COVID-19.
- Strong pairwise correlations in cause-specific excess mortality.

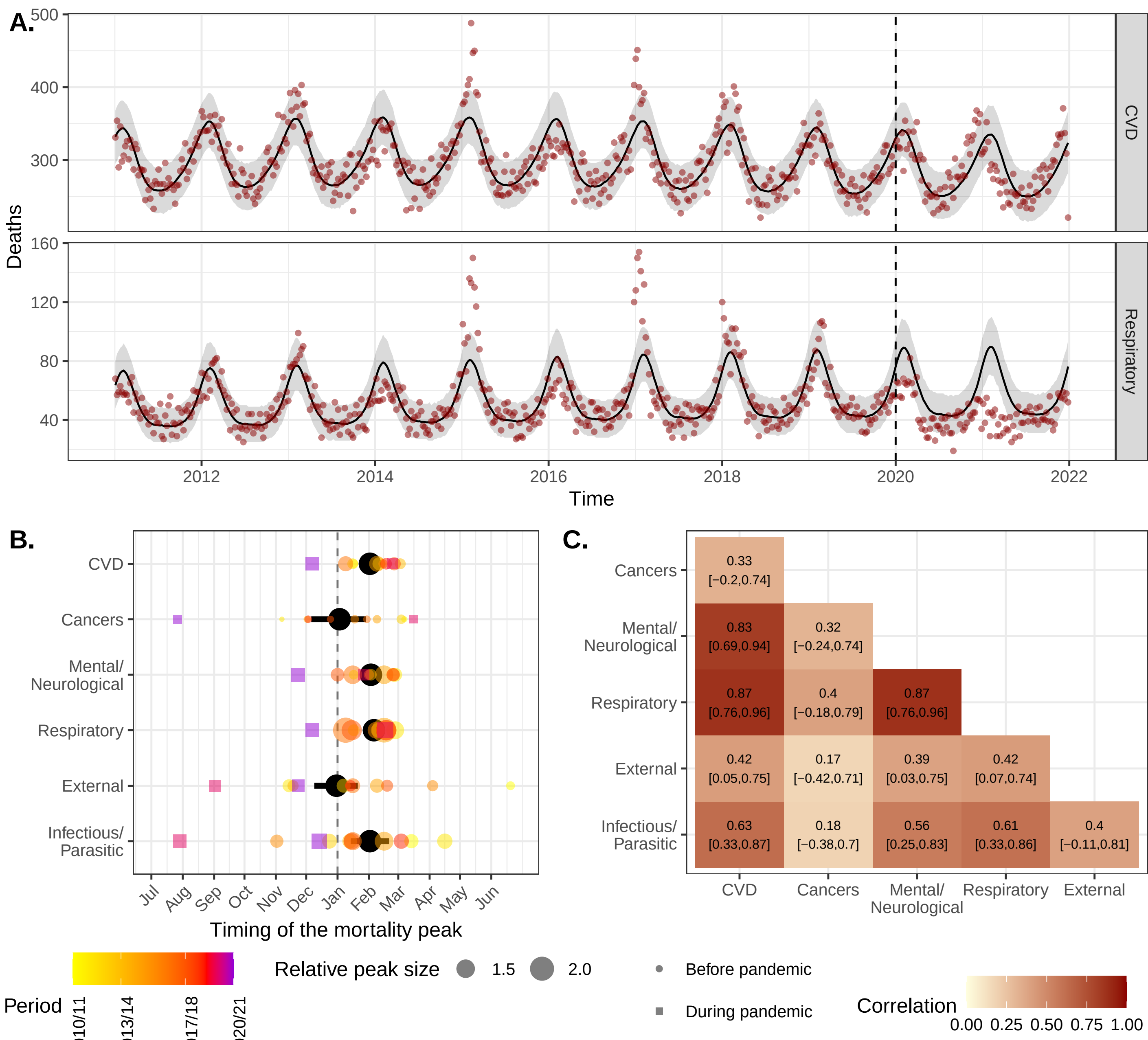


Figure 1. Model fit and cross-cause patterns for 80+. A. Weekly deaths observed (red) and model estimates/predictions (black) with 95% CrI (gray). B. Timing of mortality peaks, colored points: observed, black: posterior estimate with 95% CrI. C. Cross-cause correlation of excess mortality.

Impact of COVID-19 deaths on all-cause mortality

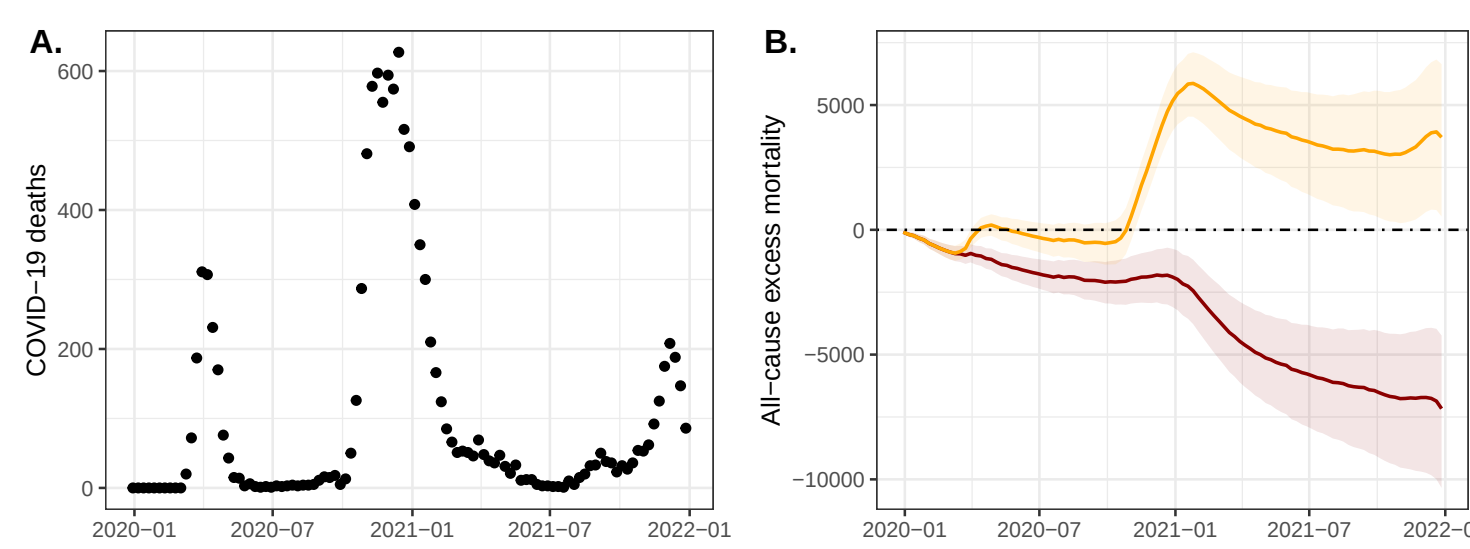


Figure 2. A. Weekly COVID-19 deaths in 80+. B. Excess mortality with and without COVID-19 in 80+.

- Sharp rise in cumulative excess mortality during major waves (see Fig 2 for 80+).
- Subsequent weeks show a mortality deficit.
- When excluding COVID-19 deaths, mortality was lower than expected in 2020–21.

Cause-specific mortality over time

Cause-specific excess by phase:

- CVD mortality excess concentrates during COVID-19 wave, followed by deficits (see Fig 3 for 80+).
- Strong mortality decrease for mental/neurological disorders (~12%), especially during the Alpha wave (~24%) in 80+.
- Cancer mortality deficit mainly during the Alpha wave in 80+.
- Respiratory (non-COVID) mortality consistently below expected across all ages, with larger deficits at older ages.

Cancer exploration

- Deficit in cancer deaths investigated using multiple-cause-of-death data.
- Total deaths with cancer did not decline in Jan–Feb 2021; slight increase in 65–79, larger in 80+.
- Largest increase observed for high-survival cancers, particularly in the 80+ age group.

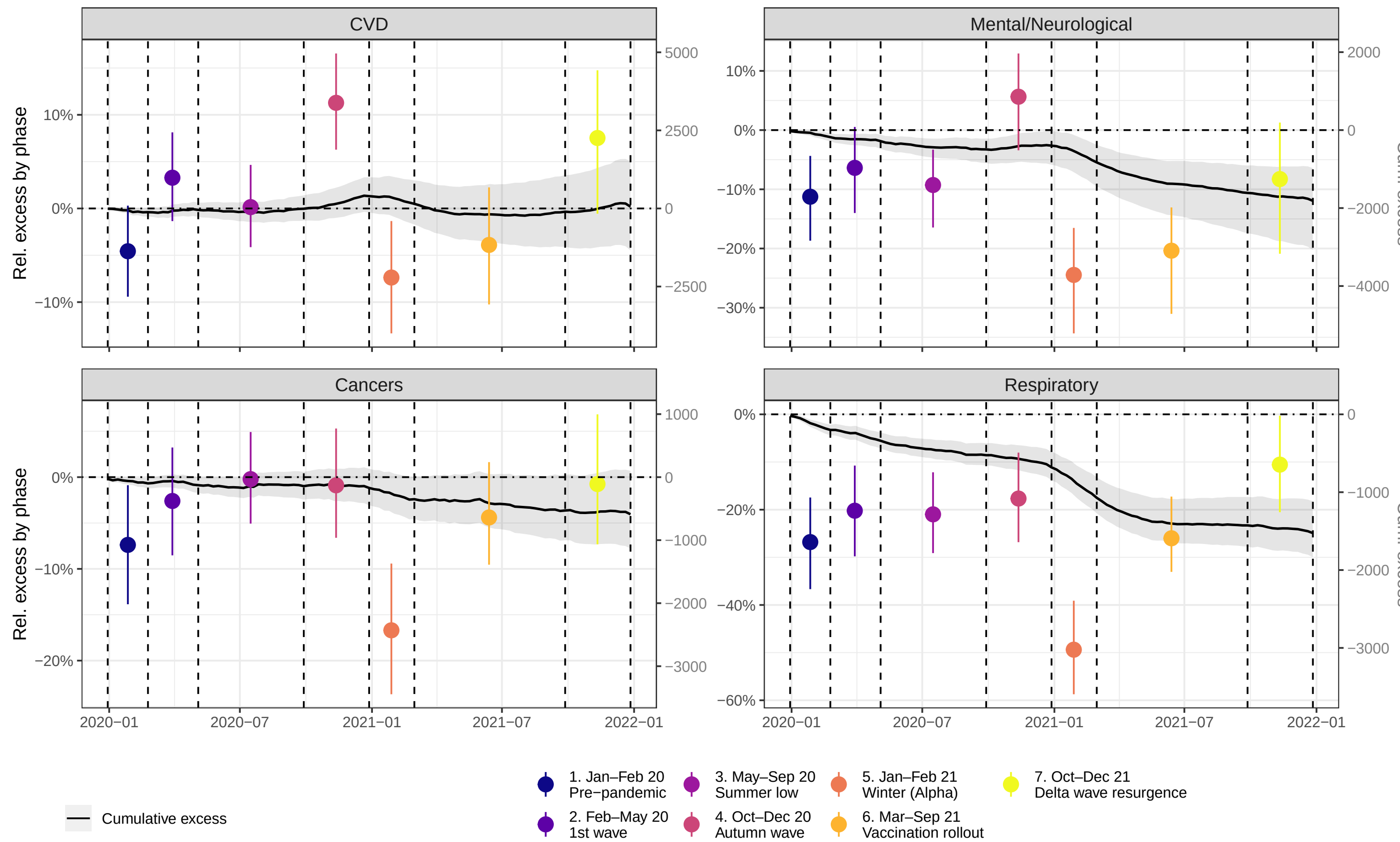


Figure 3: Cause-specific excess mortality by phase in 80+. Black lines (with gray 95% CrI) show cumulative excess mortality in 2020–21. Colored points and bars indicate phase-specific relative excess mortality (95% CrI).

Correlation between excess mortalities

- Strong positive partial correlations (up to 0.5) between COVID-19 excess mortality and excess deaths from CVD and mental/neurological diseases.
- Similar positive correlations with other respiratory diseases.
- Correlations peak at lags of 1–3 weeks, indicating COVID-19 excess deaths occurred slightly after CVD and mental/neurological deaths.
- Cancer shows weak or negative correlations with COVID-19 mortality.

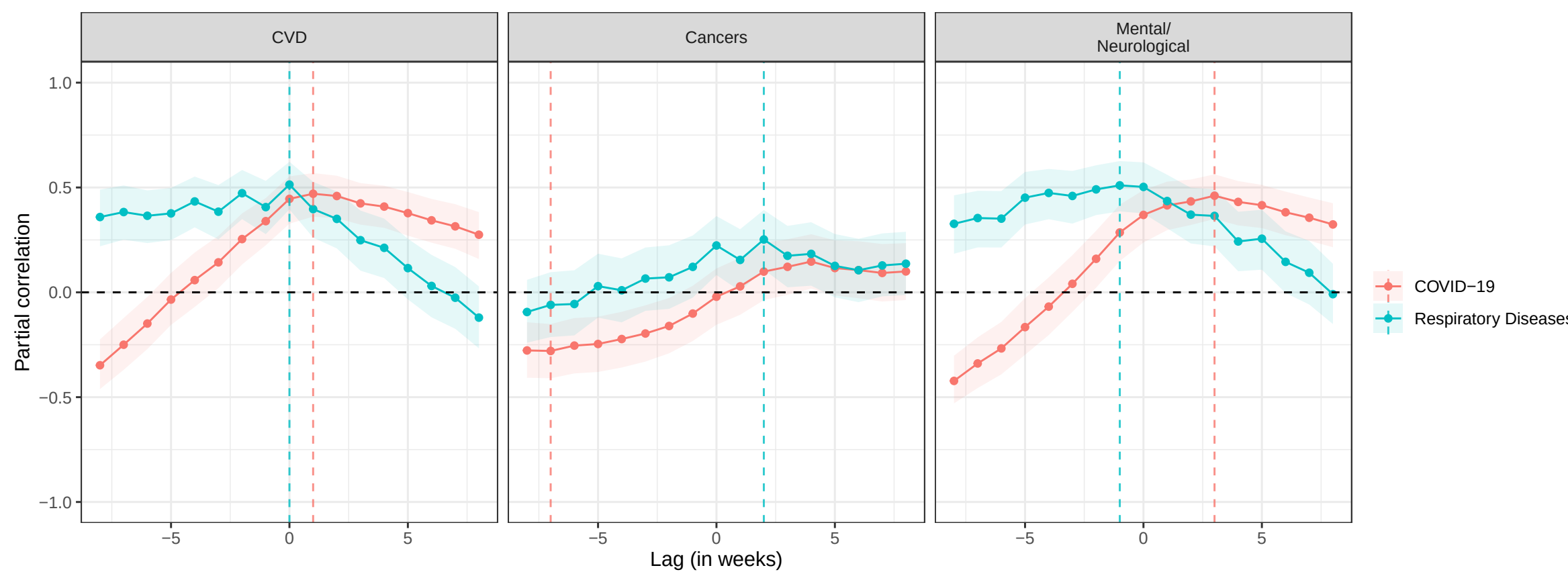


Figure 4: Partial correlation between excess mortality due to either COVID-19 (red) or to other respiratory diseases (blue) and excess mortality due to CVD, cancers or mental/neurological diseases in 80+.

Mechanisms of mortality disruption

Mechanism	Explanation	Examples
Under-ascertainment of COVID-19 deaths	COVID-19 infections not identified at the time of death inflated mortality attributed to other causes.	CVD excess during autumn 2020 aligned with COVID-19 mortality.
Delayed SARS-CoV-2 effects	Post-infection complications increased risks of death weeks to months after infection.	Low evidence for Switzerland.
Mortality displacement	Early deaths among frail individuals reduced deaths expected later, creating mortality deficits.	Deficits in dementia and CVD deaths following autumn 2020 wave. Cancer patients dying from COVID-19 instead of cancer.
Pandemic response (NPIs)	Control measures reduced circulation of respiratory pathogens and affected related causes.	Large decrease in non-COVID respiratory mortality. Lower winter mortality for dependent causes in older adults.

Table 1: Potential mechanisms driving mortality disruptions during the COVID-19 pandemic.